DIVYA PATEL

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SUMMARY

Data Science graduate student with experience spanning ML research, production deployment, and applied business analytics. Pretrained a GPT-2 model from scratch, fine-tuned transformers for NLP, designed novel computer vision architectures, and built predictive pipelines with real business impact. Proficient in Python, SQL, and R across the full ML lifecycle.

EDUCATION

Northeastern University Sep 2025 - (May 2027)
Master of Science (M.S.) in Data Science Boston, MA
GPA: 3.56/4.0
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Mining Techniques, Algorithms, Data Structures, Linear Algebra, Probability

Ahmedabad University 2021 - 2025
Bachelor of Technology (B.Tech) in Computer Science and Engineering Ahmedabad, India

TECHNICAL SKILLS

Languages: Python, SQL, R, C++, JavaScript
Backend and Data: Node.js, REST APIs, JSON, MySQL, PostgreSQL, Hadoop, Apache Spark, Linux, Git, MLflow, CUDA
Libraries: PyTorch, PyTorch Geometric, Hugging Face Transformers, Scikit-learn, XGBoost, SHAP, pandas, NumPy, OpenCV, Torchvision, Prophet, ARIMA, LSTM, GeoPandas
Tools: Jupyter Notebook, Weights and Biases, Power BI, Tableau, React.js, Claude Code
Core Competencies: Machine Learning, Deep Learning, Large Language Models (LLMs), Computer Vision, Natural Language Processing, Time-Series Forecasting, Model Interpretability, Predictive Modeling, Customer Segmentation, Business Intelligence Reporting, Automation Pipelines, Experiment Tracking

PROFESSIONAL EXPERIENCE

Data Science Intern, *Dobariya & Co. Real Estate* Aug 2024 - July 2025
Ahmedabad, India

- Built and deployed an end-to-end analytics platform on 50K+ listings using ensemble Machine Learning pipelines (XGBoost, LightGBM, Neural Network regressors), time-series forecasting (ARIMA, Prophet, LSTM), and geospatial/NLP feature engineering (GeoPandas, spaCy), achieving 92% R-squared, 15% lower RMSE, and 25% improved pricing accuracy while maintaining production reliability across daily batch runs.

Machine Learning Intern, *Augmented Systems LLP* May 2024 - July 2024
Ahmedabad, India

- Engineered a multilingual Optical Character Recognition (OCR) and NLP pipeline using Tesseract, EasyOCR and spaCy with neural translation models, achieving 95%+ text extraction accuracy, 92% translation accuracy and cutting preprocessing time by 40% across 4 regional Indian languages.
- Built a Graph Neural Network (GNN) based document similarity detector with PyTorch Geometric, reducing duplicate content processing by 30% and enabling automated multilingual text-to-English conversion workflows.

Data Analyst Intern, *Shreeya Software* May 2024 - July 2024
Ahmedabad, India

- Designed 15+ Power BI and Tableau dashboards, accelerating executive reporting speed by 30%.
- Built CRM-integrated data warehouse using star schema data modeling, boosting query efficiency by 25% and improving data reliability across business teams.

PROJECTS

Does Visual Grounding Help a Small Language Model Learn More Efficiently? (BabyLM Challenge 2026) | *Python, PyTorch, Hugging Face Transformers, GPT-2, DINOv2*

- Designed and ran a controlled pretraining experiment comparing a text-only GPT-2 model against a matched model with visual grounding, holding architecture and word budget fixed to isolate the effect of grounding alone.
- Evaluated both models on perplexity, BLiMP (67 grammatical paradigms) and EWoK, finding that grounding improved general predictive fluency but did not improve, and slightly hurt, targeted grammatical competence.

Misinformation Detection via Linguistic Style (CS6140, Spring 2026) | *Python, XGBoost, DistilBERT, SHAP, Hugging Face Transformers, spaCy*

- Fine-tuned DistilBERT on the LIAR benchmark and trained an XGBoost classifier on seven linguistic features, achieving 0.673 in-domain F1 score and 0.666 cross-domain F1 score on FakeNewsNet PolitiFact through rigorous SHAP-based evaluation methodology that surfaced a deployability gap benchmark-only testing would have hidden.

Customer Personality Analysis: Predictive Modeling for Segmentation and Spending Prediction | *Python, Scikit-learn, XGBoost, K-Means, Random Forest*

- Built a three-part predictive pipeline on retail customer data: K-Means clustering to segment customers into four actionable profiles, a Random Forest classifier to predict campaign response, and an XGBoost regression model to predict spending, translating results into concrete business recommendations.

Oriented Object Detection in Aerial Images | *PyTorch, ResNet101, DOTA Dataset, CUDA, OpenCV, Torchvision, Weights and Biases*

- Designed a novel anchor-free, single-stage detection architecture using box boundary-aware vectors (BBAVectors) for aerial imagery, achieving 75.36% detection accuracy across seven object classes with GPU-accelerated training and inference.